WEEK 1

The Deterministic Worldview and Why It Failed

18, 20 August · Objective 1

Classical physics is a total function: state in, future out, no exceptions. This week establishes that contract precisely enough that you can see what quantum mechanics breaks — because the break is narrower and stranger than the popular account suggests.

The classical contract

Newton's 1687 formulation makes a specific, testable promise. Given a system's complete state at time *t* — every position **q** and momentum **p** — the equations of motion return the state at any other time, past or future:

```
F = ma          →     d²q/dt² = F(q)/m
```

Hamilton's 1833 reformulation makes the structure explicit. Define a scalar *H*(q, p) — the total energy — and the entire dynamics falls out of two symmetric equations:

```
dq/dt =  ∂H/∂p
dp/dt = -∂H/∂q
```

This is worth dwelling on as an architectural move. Hamilton replaced *n* coupled second-order equations with 2*n* first-order ones, and in doing so identified the true state space: **phase space**, where a point (q, p) is a complete system description. Every mechanical system, from a pendulum to the solar system, became a trajectory through phase space governed by one scalar function.

> **Analogy boundary — Hamiltonian as a reducer.** *H* looks like a reducer: current state in, next state out, deterministic. The analogy holds for the type signature and fails on volume. Liouville's theorem says phase-space volume is *conserved* — classical evolution is measure-preserving and reversible. A software reducer freely collapses distinct inputs to one output. That distinction is not incidental; it is precisely what makes quantum gates unitary, and it is the reason you cannot build a quantum `if` that discards a branch.

What determinism actually claims

Laplace stated the strong form in 1814: an intellect knowing all positions and forces would find "nothing uncertain, and the future, as the past, would be present to its eyes."

Read as a systems claim, this asserts three separable properties, and it is worth keeping them apart because quantum mechanics does not treat them alike:

1. **Deterministic evolution.** The state at t_1 fixes the state at t_2.
2. **Definite values.** Every observable has a value at all times, measured or not.
3. **Non-disturbance.** Observation reads state without altering it.

Quantum mechanics keeps (1) — the Schrödinger equation is as deterministic as Hamilton's. It abandons (2) and (3). This is the single most misreported fact about the theory. The wavefunction evolves with perfect predictability; what fails is the assumption that measurement is a passive read.

> **Analogy boundary — the read/write distinction.** Classical (3) is the claim that observation is a pure read. Quantum measurement is closer to a compare-and-swap that always writes. But do not push this to "the observer's mind matters." Nothing in the formalism mentions consciousness. What matters is *irreversible correlation with an environment of many degrees of freedom* — which a photodiode accomplishes and a sleeping physicist does not.

Why it worked for three centuries

Classical mechanics was not a bad theory awaiting replacement. Le Verrier predicted Neptune's position from Uranus's orbital anomalies in 1846; Galle found it within one degree the same night. Boltzmann reduced thermodynamics to statistics with $S = k \ln W$. Maxwell unified electricity, magnetism, and light into four equations by 1865.

The framing that serves you best: classical physics is a **limit**, not an error. It is the $\hbar \to 0$ boundary of a deeper theory, the way a synchronous single-threaded model is the zero-latency limit of a distributed system. Correct inside its regime, and silently wrong outside it — which is the more dangerous failure mode, because nothing signals the crossing.

The three cracks

By 1900 three experiments resisted the framework. Each gets a chapter, but note the shared shape now:

CRACK	CLASSICAL PREDICTION	OBSERVED
Blackbody radiation	Energy → ∞ at short λ (ultraviolet catastrophe)	Finite, peaked curve
Photoelectric effect	Brighter light → faster electrons	Frequency sets energy; intensity sets count
Atomic stability	Electron spirals into nucleus in $\sim 10^{-11}$ s	Discrete spectral lines Atoms are stable

All three fail the same way: a continuum theory meets a system that is stubbornly **discrete**. Energy comes in lumps. That is the whole of the quantum hypothesis, and every subsequent strangeness follows from taking it seriously.

Why i = √−1 becomes unavoidable

Classical physics uses complex numbers as bookkeeping — a convenience for oscillations, with `Re(·)` taken at the end. Quantum mechanics cannot do this. The state itself is complex-valued, and the phase is physical.

Here is the reason, and it is an information-theoretic one. A quantum state must carry two independent quantities per branch: a magnitude that governs probability, and a phase that governs *interference between branches*. Probabilities are non-negative reals; they cannot cancel. But quantum branches must be able to cancel — that is what the double-slit dark fringes are. So amplitudes must live in a field with negative and complex values, combining linearly, with magnitudes squared to yield probabilities:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \qquad \alpha, \beta \in \mathbb{C} \qquad |\alpha|^2 + |\beta|^2 = 1$$

Two complex numbers, four real parameters, minus normalization, minus an unobservable global phase — leaves **two**. That is the surface of a sphere, and it is why the Bloch sphere in Week 7 is a sphere and not something worse.

> **Analogy boundary — amplitude as a signed weight.** Treating α as a signed weight captures cancellation and fails on rotation. A complex phase e^{iθ} is a continuous rotation in a two-dimensional plane, not a sign flip. Signs are the θ ∈ {0, π} special case; the gates in Week 7 use every angle between.

> **Try it yourself.** Open `bloch_sphere.html`. Set the state to $|0\rangle$ and apply H, then H again. You return to $|0\rangle$. Now apply H, then Z, then H — you land on $|1\rangle$. The Z did nothing observable on its own (it only changed a phase), yet it changed the final outcome completely. That is phase becoming physical through interference, and it is the mechanism every quantum algorithm exploits.

Key takeaways

- Classical mechanics promises deterministic evolution, definite values, and non-disturbing observation. Quantum mechanics keeps the first and discards the other two.
- Hamiltonian phase space identified state as (q, p) and made evolution volume-preserving — the classical ancestor of gate reversibility.
- Classical physics is the $\hbar \to 0$ limit of quantum mechanics, not a refuted theory. It fails silently at the boundary.
- All three cracks of 1900 are the same crack: continuum predictions against discrete reality.
- Complex amplitudes are forced, not chosen. Interference requires cancellation; probabilities cannot cancel; amplitudes must.

Self-check

Q1 *(Objective 1)* — Your audit pipeline is deterministic: same inputs, same hash chain. Someone claims this makes it "classical, not quantum." Using the three-part decomposition of determinism above, identify precisely which property your pipeline shares with classical mechanics and which property quantum mechanics also satisfies.

Q2 *(Objective 1)* — A colleague argues complex numbers in quantum mechanics are "just a math convenience, like phasors in AC circuit analysis." Give the specific physical observation that refutes this, and state what would have to be true of probability for the convenience view to hold.

Check your answers

A1 — Your pipeline shares **deterministic evolution** (property 1). Given identical inputs and identical prior chain state, it produces identical output — that is Hamiltonian evolution's defining feature, and quantum mechanics satisfies it too: the Schrödinger equation is fully deterministic in $|\psi\rangle$.

The interesting half is where your pipeline resembles the *quantum* side. A hash-chained audit trail deliberately violates **non-disturbance** (property 3): writing an entry changes system state irreversibly, and there is no way to record an observation without advancing the chain. You have engineered, at the application layer, precisely the property quantum measurement has at the physical layer — observation that necessarily writes. The analogy is not that your pipeline is quantum; it is that you already have working intuition for a system where "read" is not free, which most people must acquire painfully.

Property 2, definite values, is where the pipeline stays classical: your records have determinate values before you read them.

A2 — The refutation is **destructive interference**: the dark fringes of the double-slit pattern. Regions that receive particles when either slit alone is open receive *none* when both are open. Two open paths yield less than one. Since probabilities are non-negative, $P(\text{both}) \geq P(\text{one})$ always — no combination of non-negative numbers reproduces the observation. The quantities that combine must therefore support cancellation, which means they are not probabilities.

In phasor analysis the complex number is genuinely eliminable: take `Re(·)` at the end and nothing is lost, because the physical current is the real part. In quantum mechanics the elimination fails — you must square the magnitude *after* summing amplitudes, and $|\alpha + \beta|^2 \neq |\alpha|^2 + |\beta|^2$. The cross term $2 \cdot \mathrm{Re}(\alpha^* \beta)$ is the interference, and it carries the relative phase.

For the convenience view to hold, probability would have to admit negative values — which is to say it would not be probability. Attempts in this direction exist (Wigner quasi-probability distributions do go negative) and are informative precisely because that negativity is the signature of non-classicality.